

$$= \frac{1}{(\sum x_t^2)^2} (\sigma^2 \sum x_t^2 + 2\rho\sigma^2 \sum x_t x_{t-1} + 2\rho^2\sigma^2 \sum x_t x_{t-2} + \dots)$$

$$= \frac{\sigma^2}{(\sum x_t^2)} (1 + 2\rho \frac{\sum x_t x_{t-1}}{\sum x_t^2} + 2\rho^2 \frac{\sum x_t x_{t-2}}{\sum x_t^2} + \dots)$$

You should notice that the problem of autocorrelation introduces an additional expression in $Var(\hat{\beta})_{OLS}$.

Now, assume that x_t also follows an AR(1) process with $var(x_t) = \sigma_x^2$ and autocorrelation coefficient of ' r '. Since x_t is **assumed to be stochastic** (see Unit 10 of this course), we take the asymptotic variance of $\hat{\beta}$ in the presence of AR(1) process. $Var(\hat{\beta})_{AR(1)}$ is given by,

$$Var(\hat{\beta})_{AR(1)} = E\left(\frac{\sum x_t u_t}{\sum x_t^2}\right)^2$$

$$= \frac{\sigma^2}{(\sum x_t^2)} (1 + 2\rho r + 2\rho^2 r^2 + \dots) = \frac{\sigma^2}{T\sigma_x^2} \frac{(1 + \rho r)}{(1 - \rho r)} \quad \dots(9.11)$$

(T = number of observations)

Hence, we can write

$$Var(\hat{\beta})_{AR(1)} = Var(\hat{\beta})_{OLS} \frac{(1 + \rho r)}{(1 - \rho r)} \quad \dots (9.12)$$

Equation (9.12) has the following implication:

$Var(\hat{\beta})_{OLS}$ may underestimate the $Var(\hat{\beta})_{AR(1)}$ in the presence of autocorrelation. The nature of bias however depends on the values taken by ρ and r . If ρ and r are both positive, then,

$$Var(\hat{\beta})_{OLS} < Var(\hat{\beta})_{AR(1)}$$

From (9.12) we observe that in the presence of AR(1) process, the formula for calculation of variance of the OLS estimator $\hat{\beta}$ is

$$Var(\hat{\beta})_{AR(1)} = \frac{\sigma^2}{T\sigma_x^2} \frac{(1 + \rho r)}{(1 - \rho r)} \quad \dots(9.13)$$

In such a case, using the usual OLS formulae for computation of variance of OLS estimators leads to biased results. Moreover, depending on the values of ρ and r , the variance of the OLS estimator $Var(\hat{\beta})_{AR(1)}$ may be found to be higher than what it would be in the absence of autocorrelation. Hence, the presence of autocorrelation has a tendency to inflate the standard errors.

Application of OLS may give misleading standard errors of estimators in the presence of autocorrelation. In this case, Generalised Least Squares (GLS) should be used as it gives estimators which have minimum variance. When we allow for autocorrelation and use $Var(\hat{\beta})_{AR(1)}$, wider confidence intervals are obtained as

compared to those obtained from GLS procedure. Thus, the likelihood of accepting the null hypothesis increases and the coefficient may be declared insignificant despite being significantly different from zero.

9.3.3 Biasedness of $\text{Var}(\hat{\beta})_{OLS}$

$$\text{Var}(u_t) = \sigma^2$$

$$\text{and, } \text{Var}(\hat{\beta})_{OLS} = \frac{\sigma^2}{(\sum x_t^2)}$$

For unbiasedness of $\text{Var}(\hat{\beta})_{OLS}$, we are required to prove that,

$$E[\text{Var}(\hat{\beta})_{OLS}] = \text{Var}(\hat{\beta})_{OLS}$$

For the above to hold true, $E(\hat{\sigma}^2)$ should equal σ^2 . For a two variable regression model,

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_t^2}{T-1} \quad \dots (9.14)$$

$$E(\sum \hat{u}_t^2) = E(\sum u_t^2) + E[(\hat{\beta} - \beta)^2 \sum x_t^2] - 2E[(\hat{\beta} - \beta) \sum x_t u_t]$$

$$\begin{aligned} \text{[Since } \hat{u}_t &= y_t - \hat{y}_t = \beta x_t + u_t - \hat{\beta} x_t = u_t + \beta x_t - \hat{\beta} x_t \\ &= u_t - x_t (\hat{\beta} - \beta)] \end{aligned}$$

$$\text{Or, } E(\sum \hat{u}_t^2) = T\sigma^2 + E\left(\frac{\sum x_t u_t}{\sum x_t^2}\right)^2 \sum x_t^2 - 2E\left(\frac{\sum x_t u_t}{\sum x_t^2}\right) \sum x_t u_t$$

$$\text{Or, } E(\sum \hat{u}_t^2) = T\sigma^2 + E\frac{(\sum x_t u_t)^2}{\sum x_t^2} - 2E\frac{(\sum x_t u_t)^2}{\sum x_t^2}$$

$$\text{Or, } E(\sum \hat{u}_t^2) = T\sigma^2 + \frac{\sigma^2(1+r\rho)}{(1-r\rho)} - 2\frac{\sigma^2(1+r\rho)}{(1-r\rho)}$$

(Refer to equations (9.10) and (9.11))

$$\text{Or, } E(\sum \hat{u}_t^2) = \sigma^2 \left[T - \frac{(1+r\rho)}{(1-r\rho)} \right] \quad \dots (9.15)$$

In classical regression model we have seen that $E\left(\frac{\sum \hat{u}_t^2}{T-1}\right) = E(\hat{\sigma}^2)$. Here in the AR(1) model we find that $E(\sum \hat{u}_t^2) = \sigma^2 \left[T - \frac{(1+r\rho)}{(1-r\rho)} \right]$. Thus in the case of AR(1), we have

$$E(\hat{\sigma}^2) \neq \sigma^2 \quad \dots (9.16)$$

We cannot say whether the bias is upward or downward; it depends upon the sign of 'r' and 'ρ'. If both 'r' and 'ρ' are positive, then $E(\hat{\sigma}^2) < \sigma^2$ and the bias is downward.

Check Your Progress 1

- 1) “If the OLS method is applied to time series data, we get biased estimates as the errors are autocorrelated”. True or False? Explain.

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- 2) “Autocorrelation in error terms caused biased standard errors.” True or False? Explain.

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- 3) For an AR(2) process, show that $cov(u_t, u_{t-2}) = \rho^2 \frac{\sigma_\varepsilon^2}{(1-\rho^2)}$

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9.4 DETECTION OF AUTOCORRELATION

In the previous section we discussed the consequences of autocorrelation. In this section we discuss some of the methods to detect autocorrelation.

9.4.1 Residual Plot

The actual errors in a regression model are not known to us. What we know are the estimated values of the error terms, i.e., residuals. By plotting the residuals ‘ $\hat{u}$ ’ against time ‘ t ’, we can observe if there is a systematic pattern. If no systematic pattern is observed, it indicates no autocorrelation. If, however, the residuals show a linear downward or upward trend, or a cyclical or quadratic pattern, then it indicates the presence of autocorrelation.

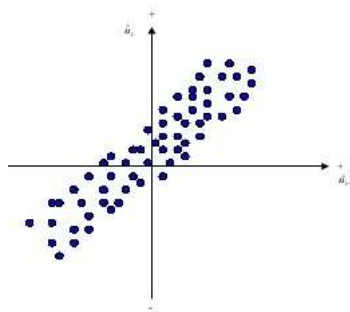


Fig.9.1: Positive Autocorrelation

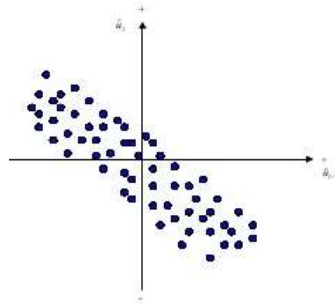


Fig. 9.2: Negative Autocorrelation

Alternatively, we can plot $\hat{u}_t$ against $\hat{u}_{t-1}$ as shown in Fig. 9.1 and Fig. 9.2. The residual plot method tests for the AR(1) process. Fig. 9.1 shows positive autocorrelation while Fig. 9.2 displays negative autocorrelation. A major limitation of the residual plot is that its interpretation is highly subjective. In view of this, we look for quantitative tests for autocorrelation.

Some of the quantitative tests are discussed below.

9.4.2 Durbin-Watson (DW) Test

A popular test for autocorrelation is the Durbin-Watson (DW) test. It is given by

$$d = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2} \quad \dots(9.17)$$

It is the ratio of the sum of squared differences in successive residuals to the overall residual sum of squares (RSS). It can be further elaborated as

$$d = \frac{\sum \hat{u}_t^2 + \sum \hat{u}_{t-1}^2 - 2 \sum \hat{u}_t \hat{u}_{t-1}}{\sum \hat{u}_t^2}$$

There are $(T - 1)$ observations in the numerator as taking successive differences results in loss of one observation. For large size samples, $\sum \hat{u}_t^2$ and $\sum \hat{u}_{t-1}^2$ are nearly equal.

$$\text{Hence, } d = 2 - \frac{2 \sum \hat{u}_t \hat{u}_{t-1}}{\sum \hat{u}_t^2} \approx 2(1 - \hat{\rho}) \quad \dots(9.18)$$

If there is no autocorrelation in a regression model, then $\hat{\rho} = 0$. By applying this value in (9.18) we find that $d = 2$. On the other hand, If $\hat{\rho} = +1$, $d = 0$, and if $\hat{\rho} = -1$, $d = 4$. Thus, d being close to zero implies positive autocorrelation, whereas d being close to 4 implies negative autocorrelation. Thus, d lies between 0 and 4, i.e., $0 \leq d \leq 4$. We can set our null hypothesis of no autocorrelation as $H_0: d = 2$. If d is found to be around 2, we accept the H_0 ; if it exceeds the critical value we reject the H_0 .

Somehow the d -statistic does not have a particular critical value for acceptance or rejection of the null hypothesis of no autocorrelation. However, it reports the lower and upper limits d_L and d_U respectively. These limits are reported for the number of observations in the sample and the number of independent variables (excluding the constant term) in the regression model.

If the value of d lies between 0 and d_L , there is positive autocorrelation. If $4 - d_L < d < 4$, there is negative autocorrelation. If $d_U < d < 4 - d_U$, there is no autocorrelation. If the value of d lies between d_L and d_U ($d_L \leq d \leq d_U$ or $4 - d_U \leq d \leq 4 - d_L$), this test is inconclusive as this is the zone of indecision (see Fig. 9.3). This is an important limitation of the Durbin-Watson test. Typically, it is observed that as sample size increases, the zone of indecision shrinks.

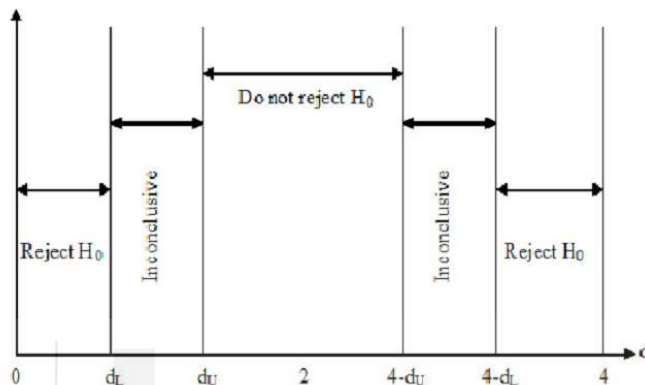


Fig. 9.3: Rejection Zones under the DW Test

The DW statistic is based on certain assumptions: i) intercept term is present in the regression model, ii) the independent variables are fixed in repeated sampling (maintaining this assumption is difficult in case of time series data), iii) the error terms are normally distributed, and iv) the error terms follow AR(1) process. Hence, the d -statistic may be unreliable if the error terms are not NIID (normally, independently and identically distributed). However, in large samples, the d -statistic follows a standard normal distribution. Further, this test can be used only in the case of first order autocorrelation. The DW test cannot be used if we have lagged dependent variables in the regression model.

9.4.3 Breusch-Godfrey (BG) Test

The BG test overcomes the limitations of the DW test. It is a general test for higher order autocorrelations. It can be applied in models having lagged values of dependent variable as independent variables. It can also be applied in case of moving averages of white noise error terms. This test is derived from the Lagrange Multiplier (LM) principle. Hence, it is also known as LM test.

Let the regression model be specified as follows:

$$y_t = \sum_{i=1}^k x_{it} \beta_i + u_t; t = 1, 2, \dots, n \quad \dots (9.19)$$

Here, lagged dependent variables can also be included as independent variables in the x 's.

Let u_t follow an AR(p) process such that

$$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \dots + \rho_p u_{t-p} + \varepsilon_t \quad \dots (9.20)$$

In the above equation,

$$\varepsilon_t \sim IN(0, \sigma^2)$$

Now we specify the null hypothesis as

$$H_0: \rho_1 = \rho_2 = \dots = \rho_p = 0 \quad \dots(9.21)$$

The BG Test involves the following steps:

- (i) Use OLS to estimate the primary regression equation $y_t = \sum_{i=1}^k x_{it} \beta_i + u_t$ and get $\hat{u}_t$.
- (ii) Regress $\hat{u}_t$ on the original values of x_{it} (as well as any other variable used as regressor) and lagged values of the residuals (for example, if $p = 3$, we will include three lagged values of the estimated residuals as regressors). The auxiliary regression equation is as follows:

$$\hat{u}_t = \sum_{i=1}^k x_{it} \gamma_i + \sum_{i=1}^p \hat{u}_{t-i} \rho_i + \tau_t \quad \dots(9.22)$$

- (iii) If the sample size is large, then Breusch and Godfrey have shown that

$$(T - p)R^2 \sim \chi_p^2 \quad \dots(9.23)$$

In equation (9.23), T is the sample size, p is the number of autoregressive parameters in the auxiliary regression, and R^2 is the coefficient of determination of the auxiliary regression equation. BG test is asymptotically (it means, as sample size increases infinitely) distributed as χ^2 with p degrees of freedom.

- (iv) Once we get the computed value of the χ^2 in (9.23) we can compare it with the standard value (tabulated value). If computed value exceeds tabulated value we reject the null hypothesis.

Check Your Progress 2

- 1) Explain the steps followed the Durbin Watson test.

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- 2) What are the limitations of the DW test? Can you suggest some alternative tests which can overcome these limitations?

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- 3) Consider the model $y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + u_t$. The DW statistic is 0.85. The number of observations is 80. Also given that $d_L = 1.41$ and $d_U = 1.50$. Is there a problem of autocorrelation in the data?

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9.5 REMEDIAL MEASURES

If the DW statistic is found to be statistically significant, it is not necessarily due to AR(1) errors. It can be due to the impact of autocorrelated omitted variables or due to AR(2) or moving average errors. In this case, the natural question which arises is: what is the appropriate course of action if DW statistic is significant? First, we should check if the correlation in the error terms is pure autocorrelation or it is due to some omitted variables/ model mis-specification. If there is pure autocorrelation, then we can appropriately transform the original model. The method of Generalised Least Squares (to be discussed in Unit 11) can be employed to correct pure autocorrelation. For large samples, the Newey West Method can be employed. The various remedial measures are described below.

1. Quasi-first Differencing

Given, $y_t = \alpha + \beta x_t + u_t$; $t = 1, 2, \dots, T$

We can obtain $y_{t-1} = \alpha + \beta x_{t-1} + u_{t-1}$

Now, $\rho y_{t-1} = \alpha\rho + \beta\rho x_{t-1} + \rho u_{t-1}$

Taking the difference with respect to $y_t = \alpha + \beta x_t + u_t$, we get,

$$y_t - \rho y_{t-1} = \alpha(1 - \rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t \quad \dots(9.24)$$

[Errors follow AR(1) process such that $u_t = \rho u_{t-1} + \varepsilon_t$]

Equation (9.24) is known as Generalised or Quasi-Difference equation. The error terms ε_t satisfies the usual classical assumptions (independent having constant variance σ_e^2). Now, the OLS procedure can be applied by regressing the transformed variable $y_t^* = (y_t - \rho y_{t-1})$ on $x_t^* = (x_t - \rho x_{t-1})$. The resulting estimators are best, linear and unbiased (BLUE). In taking differences, one observation is lost; hence, $(T - 1)$ observations are used. In order to compensate for loss of one observation, **Prais-Winsten transformation** is used wherein the first observation on y and x are transformed such that $y_1^* = y_1\sqrt{1 - \rho^2}$ and $x_1^* = x_1\sqrt{1 - \rho^2}$. Effectively, this procedure is the GLS procedure. The GLS procedure has T observations which includes y_1^* and x_1^* .

Applying OLS to the transformed model which satisfies the classical assumptions is GLS. In this method, the variables are transformed by incorporating the autocorrelation parameter ' ρ '. The GLS procedure produces efficient estimates since all available information (about autocorrelation or heteroscedasticity) is used. In the presence of autocorrelation, the GLS estimator is the best linear unbiased estimator (BLUE).

$$\hat{\beta}_{GLS} = \frac{\sum_{t=2}^n (x_t - \rho x_{t-1})(y_t - \rho y_{t-1})}{\sum_{t=2}^n (x_t - \rho x_{t-1})^2} + C \quad \dots (9.25)$$

$$Var\hat{\beta}_{GLS} = \frac{\sigma^2}{\sum_{t=2}^n (x_t - \rho x_{t-1})^2} + D \quad \dots(9.26)$$

(C and D are correction factors)

One shortcoming of the quasi-differencing method is that ρ should be known. However, ρ is seldom known. There are various methods of estimating ρ which are discussed in Section 9.6. A limitation of using $\hat{\rho}$ is that it can be subject to sampling error. An alternative to quasi-differencing is the first-differencing method.

2. First-Differencing

Recall the quasi-difference equation given at (9.24)

$$y_t - \rho y_{t-1} = \alpha(1 - \rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t$$

If $\rho = +1$, then the above equation reduces to a first difference equation,

$$y_t - y_{t-1} = \beta(x_t - x_{t-1}) + (u_t - u_{t-1})$$

$$\text{Or, } y_t - y_{t-1} = \beta(x_t - x_{t-1}) + \varepsilon_t \quad \dots(9.27)$$

As mentioned earlier, the error term ε_t follows classical assumptions of the stochastic error term. Although u_t and u_{t-1} are correlated, the first difference of the error terms ($u_t - u_{t-1}$) are not correlated. Now, we can regress ($y_t - y_{t-1}$) on ($x_t - x_{t-1}$) by OLS method *without including the constant term* in the model. If there is a linear trend term in the regression equation (for example, $y_t = \alpha + \delta t + \beta x_t + u_t$), then the first difference equation has a constant term:

$$(y_t - y_{t-1}) = \delta + \beta(x_t - x_{t-1}) + (u_t - u_{t-1}) \quad \dots(9.28)$$

Essentially, the first difference equation may be estimated if the Durbin-Watson statistic is small (rule of thumb: $d < R^2$). In a strict sense, first difference transformation is valid only if $\rho = +1$. In order to test whether $\rho = +1$ we use the Berenblut-Webb test.

The first difference transformation induces stationarity in the time series. If $\rho = +1$, then the variance $Var(u_t) = \frac{\sigma_\varepsilon^2}{(1-\rho^2)}$ and $Cov(u_t, u_{t-s}) = \rho^s \frac{\sigma_\varepsilon^2}{(1-\rho^2)}$ become infinite. Hence, the series u_t becomes non-stationary. With first difference transformation, the error term is ε_t , which does not have such problems.

Note that the R^2 of the levels equation (say, $y_t = \alpha + \beta x_t + u_t$) and the first difference equation ($(y_t - y_{t-1}) = \delta + \beta(x_t - x_{t-1}) + (u_t - u_{t-1})$) cannot be compared because they have different dependent variables. However, the Residual Sum of Squares (RSS) can be compared after making some adjustments.

The steps to be followed are as follows:

- (i) Estimate the RSS from the first difference equation as well as the levels equation.
- (ii) Multiply the RSS from the levels equation by $\frac{2(n-k-1)(1-\rho)}{(n-k)}$ (or, $\frac{(n-k-1)d}{(n-k)}$, which can be derived from the above)

- (iii) Use the estimated ρ from the levels equation, that is, $\hat{\rho}$, [or, the DW statistic, $d \approx 2(1 - \hat{\rho})$]. Here, k is the number of regressors.
- (iv) Now, the RSS from the first difference equation and the levels equation can be compared.

Recall that the variance of the error term $\text{var}(u_t) = \text{RSS}/\text{degrees of freedom}$. Therefore, variance of the error term in first difference equation

$$\begin{aligned} &= \text{Var}(u_t - u_{t-1}) \\ &= \text{Var}(u_t) + \text{Var}(u_{t-1}) - 2 \text{Cov}(u_t, u_{t-1}) \\ &= \sigma^2 + \sigma^2 - 2\sigma^2\rho = 2(1 - \rho)\sigma^2 \end{aligned}$$

You should note that the above arguments do not hold in the presence of lagged dependent variables among the explanatory variables. Hence, this discussion excludes them by assumption. Having obtained comparable RSS, comparable R^2 can be obtained as well. Using $\text{RSS} = S_{yy}(1 - R^2)$, we get,

$$\frac{(1 - R_D^2)}{(1 - R_1^2)} = \left(\frac{\text{RSS}_0}{\text{RSS}_1} \right) \frac{(n-k-1)d}{(n-k)} \quad \dots(9.29)$$

$$\text{Here, } S_{yy} = \sum y_i^2 - \bar{y}^2$$

$R_1^2 = R^2$ from the first difference equation

$R_D^2 =$ comparable R^2 from the levels equation (i.e., without differencing)

$\text{RSS}_0 = \text{RSS}$ from the levels equation

$\text{RSS}_1 = \text{RSS}$ from the first difference equation

Sometimes, we get high R^2 in levels form equation and low R^2 in first difference equation. But the DW statistic may be found to be low in the levels form equation. This often suggests that the equation is mis-specified. In this case, we should use the first difference equation. Low R^2 in first difference equation may indicate that the dependent and independent variables are unrelated.

It should also be noted that the above discussion on first differencing and quasi-first differencing assumes a particular correlation structure between u_t and u_{t-1} . Sometimes, u_t can be found to be correlated with u_{t-4} . The DW statistic will not be able to detect this, and it has to be modified as follows:

$$d_4 = \frac{\sum (\hat{u}_t - \hat{u}_{t-4})^2}{\sum \hat{u}_t^2} \quad \dots (9.30)$$

In this case, fourth differencing or quasi-fourth differencing will have to be used, that is, regress $y_t - \hat{\rho}y_{t-4}$ on $x_t - \hat{\rho}x_{t-4}$, where, $\hat{\rho}$ is the correlation coefficient between $\hat{u}_t$ and $\hat{u}_{t-4}$.

3. The Newey-West Method

Under this method, the OLS method is used but the standard errors are corrected for autocorrelation. For large samples, the Newey-West method gives autocorrelation corrected standard errors of estimators. These are called Newey-West standard errors. In fact, these standard errors are corrected for both autocorrelation and heteroscedasticity and are also known as HAC (Heteroscedasticity-and-Autocorrelation-Corrected Standard Errors). This is an extension of White's Heteroscedasticity-Corrected standard errors. The Newey-West method is valid for large samples only.

9.6 METHODS OF ESTIMATING ρ

If ' ρ ' is not close to 1 as required for First-Differencing method, it can be estimated using some of the methods described below.

1. From Durbin-Watson Statistic

The Durbin-Watson statistic $d \approx 2(1 - \hat{\rho})$. Hence,

$$\hat{\rho} \approx 1 - \frac{d}{2} \quad \dots(9.31)$$

However, this method can be used in large samples only.

2. From the Residuals

Errors follow AR(1) process such that $u_t = \rho u_{t-1} + \varepsilon_t$. Estimate ρ by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ from the regression,

$$\hat{u}_t = \rho \hat{u}_{t-1} + v_t \quad \dots (9.32)$$

(v_t is the white noise).

3. Iterative Procedure

Taking some starting value of ρ , this method estimates $\hat{\rho}$ by successive approximation. Some iterative methods are discussed below:

3a. Cochrane–Orcutt Procedure: This method estimates $y_t = \alpha + \beta x_t + u_t$ by OLS and the residuals $\hat{u}_t$ are obtained. Hence, $\hat{\rho}$ which is the first-order coefficient of autocorrelation is obtained as follows:

$$\hat{\rho} = \sum \hat{u}_t \hat{u}_{t-1} / \sum \hat{u}_t^2 \quad \dots (9.33)$$

This method can also be applied in the case of higher order autoregressive process.

3b. Durbin's Procedure: Rewrite the equation,

$$y_t - \rho y_{t-1} = \alpha(1 - \rho) + \beta(x_t - \rho x_{t-1}) + \varepsilon_t$$

$$\text{as } y_t = \alpha(1 - \rho) + \rho y_{t-1} + \beta x_t - \rho \beta x_{t-1} + \varepsilon_t \quad \dots (9.34)$$

Now, regressing y_t on y_{t-1} , x_t and x_{t-1} , the coefficient of y_{t-1} gives an estimate of ρ . Using this estimate of ρ , find $(y_t - \rho y_{t-1})$ and $(x_t - \rho x_{t-1})$ and regress $(y_t - \rho y_{t-1})$ on $(x_t - \rho x_{t-1})$ to get an estimate of β . The constant term in this

regression corresponds to $\alpha(1 - \rho)$. Since the estimate of ρ is known, α can be obtained. Hence we get α and β , the estimates of the parameters of the original equation $y_t = \alpha + \beta x_t + u_t$.

A disadvantage of using this method is that it involves a number of variables in the regression. This is more of a problem if there are several independent variables.

4. Grid-Search Procedures

4a. Hidreth and Lu Procedure: Let $y_t^* = y_t - \rho y_{t-1}$ and $x_t^* = x_t - \rho x_{t-1}$. Assuming different values of ρ in the range +1 to -1 at intervals of 0.1, obtain y_t^* on x_t^* for different values of ρ and regress y_t^* on x_t^* . From each such regressions, obtain the RSS. Now, choose the regression having minimum RSS and choose the value of ρ corresponding to it. Smaller intervals say, 0.01 can be taken around the chosen value of ρ and this procedure can be repeated.

4b. Maximum Likelihood Procedure: If the error terms ε_t are normally distributed, the log likelihood function can be written as follows:

$$\log L = \text{constant} - \frac{T}{2} \log \sigma_e^2 + \frac{1}{2} \log(1 - \rho^2) - \frac{Q}{2\sigma_e^2} \quad \dots(9.35)$$

$$\text{Here, } Q = \sum [y_t - \rho y_{t-1} - \alpha(1 - \rho) - \beta(x_t - \rho x_{t-1})]^2 \quad \dots(9.36)$$

To get the ML estimates, we use the grid-search procedure. For each value of ρ , we compute the RSS (ρ) and choose that value of ρ for which $[\frac{T}{2} \log RSS(\rho) - \frac{1}{2} \log(1 - \rho^2)]$ is the minimum.

In conclusion, we can say that estimating ' ρ ' by these different methods does not make much difference in large samples, because the OLS estimators are consistent even in the presence of autocorrelation. Using any of these methods, we get consistent estimates of the true ' ρ '. Having obtained an estimate of ' ρ ', it is used to estimate GLS. Since ' $\hat{\rho}$ ' is used, it is often known as *estimated GLS* (EGLS) or *feasible GLS* (FGLS)).

In small samples, however, the estimated coefficients (parameters of the transformed model) may not be BLUE because ' $\hat{\rho}$ ' is used instead of true ' ρ '. The estimated OLS coefficients display usual optimum properties asymptotically (when sample size increases infinitely). Moreover, in small samples, it is sometimes better to continue using OLS. As a rule of thumb, if sample size is 20 and $\hat{\rho} < 0.3$, OLS is better than FGLS ($\hat{\rho}$ is the estimated first order coefficient of autocorrelation). Additionally, it must be noted that this discussion essentially assumes that the error terms are AR(1). If they are higher order autoregressive, it is sometimes advised to continue assuming AR(1) process rather than ignoring autocorrelation altogether and using usual OLS. The issue is debatable, and some researchers contradict this by saying that doing so may worsen the problem.

Check Your Progress 3

- 1) Consider the model $y_t = \alpha + \beta x_t + u_t$

$$u_t = \rho u_{t-1} + \varepsilon_t; 0 \leq \rho \leq 1$$

$$\varepsilon_t \sim IN(0, \sigma^2)$$

By regressing Δy_t on Δx_t , is it possible to get more efficient estimates of β than by regressing y_t on x_t ?

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- 2) Given $y_t = 2.8 + 0.5 x_t + 0.8 y_{t-1}$
 (0.38) (0.06)

The p-values are given in parenthesis. $R^2 = 0.95$. $DW = 1.89$. Here, R^2 is high and DW is also closer to 2. Identify the problem in this equation.

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9.7 LET US SUM UP

Autocorrelation is a typical problem in time series data. Even with autocorrelated errors, it is seen that the OLS estimators are unbiased. However, they no longer have minimum variances and the variances are also biased. Thus, the usual tests of significance invalid in the presence of autocorrelation among error terms. Of the various tests to detect autocorrelations, the DW test is the most popular. But, even a statistically significant DW statistic does not necessarily mean that the errors are autocorrelated. Sometimes incorrect specification of the model or omission of relevant variables may cause the DW statistic to be statistically significant. Thus, one should first verify if there is pure autocorrelation or not before proceeding for correcting it. There are other problems with the DW statistic. It cannot detect higher order autocorrelations and it cannot be applied to models having lagged explained variables as independent variables. In view of these shortcomings, the BG test scores over the DW test. Having detected the presence of pure autocorrelation, the next step is to correct for it. The method of GLS can be used but one has to estimate ρ for it. If ρ is close to 1, then first-difference transformation can be applied. There are various methods of estimating ' ρ '. In large samples, any method of estimating ρ provides consistent estimates of true ρ . This is because despite autocorrelation, the OLS estimators are consistent, that is, they tend towards their true parameter values with increase in sample size. However, in small samples, the results have to be interpreted cautiously. The estimated OLS coefficients (parameters of the transformed model) have the optimum properties but only in large samples.